On Acceptable Modification Limits of Electroacoustic Speech Signals for Data Hiding

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Abstract – The paper focuses on crypto security for data hiding in electroacoustic speech signals, and an example of data hiding in a speech signal based on segment fundamental frequency modification is described. In the process of embedding of data into speech artificial distortions can be perceivable for human ear. The author sets a task to determine acceptable limits of modification to maintain optimal security level and natural speech communication.

Data hiding, steganography, quantization index modulation.

I. INTRODUCTION

Methods that embed data into such continuous value of bearing parameters, for example phase coding, spread spectrum, echo data hiding, data hiding on the basis of the modification of fundamental frequency and duration of voiced speech segments are of great value for systems of secret communication. These methods keep embedded data immune to noise, filtering, lossy compression, vocoder, analog-to-digital and digital-to-analog converting; the original host audio signal is not used to extract the embedded data [1].

In order to embed discrete data into continuous value of speech signal time and level quantization is required. Signal segmentation into natural homogeneous voiced stationary parts is analogous to time quantization, and for level quantization the coding method with quantization index modulation (QIM) is the most simple for stream oriented implementation. [2].

II. MATERIALS AND METHODS

The physical result of QIM coder transformation of fundamental frequency, for example, in the transmitting side is that segments with standardized pitches corresponding centers of quantization intervals are formed in the output from natural speech segments of arbitrary fundamental frequency that enter the input. In the input the decoder extracts embedded data on the basis of comparison of estimated value of pitches of the received signal with the coding table common for the sender and the receiver. [3].

Finding standardized pitches (standardized values of echo signal, phase or duration of voiced speech segments) intruders can evidently easily determine the availability of hidden data in the transmitting channel. They can even read them in case if they represent a message and are transmitted as an open text in one of known coding systems. In case if the message is ciphered, the crypto analyst deals with it. Stream oriented ciphers are considered to be less immune to deciphering than block ciphers, so it means that the crypto analyst also has a chance. But the most important is that the stego channel is identified. Depending on the results of crypto analysis and the intruder's objective he can continue tapping or influence the channel in order to destroy the embedded message or intrude to the receiver false information, i.e. turn into an active intruder.

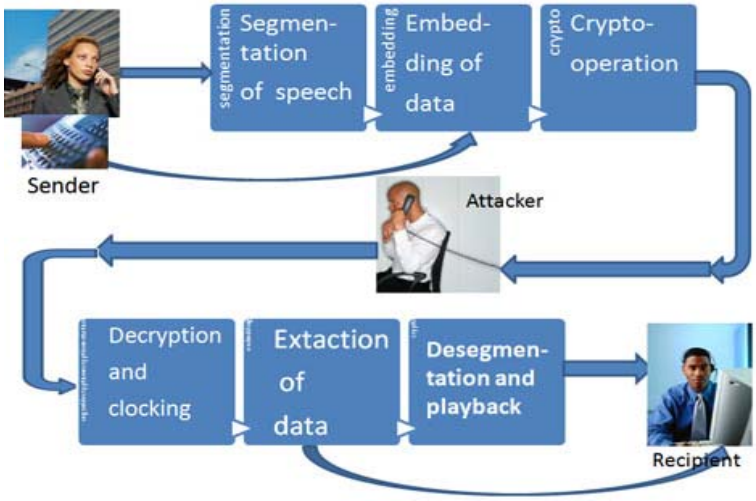


Figure 1 Algorithm of data hiding by means of QIM modification and crypto covered transmission in open channel

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TABLE I. FUNDAMENTAL FREQUENCY MODIFICATION FOR DATA HIDING

	Segment No	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
1	Fundamental frequency of empty container	*114*	*114*	*114*	*130*	*206*	*115*	*115*	*159*	*206*
2	Stego codes of empty container	14	14	14	30	06	15	15	59	06
3	Stego enclosure (simb/ binary/ decimal)	**Rus/** 00000/0	**К/** 11110/30	**О/** 00011/03	**М/** 00111/39	**П/** 01101/13	**А/** 11000/24	**Н/** 00110/06	**И/** 01100/44	**Я/** 1110/ 29
4	Fundamental frequency of filled container	*100*	*130*	*103*	*139*	*213*	*124*	*106*	*144*	*229*
5	Gamma (simb/binary)	**Lat/** 11111	**E/** 10000	**M/** 00111	**B/** 10011	**E/** 10000	**D/** 10010	**I/** 01100	**N/** 00110	**G/** 01011
6	Ciphered text (simb/binary/decimal)	**Lat/** 11111/31	**C/** 01110/14	**Space mark/** 00100/04	**S/** 10100/20	**Q/** 11101/18	**R/** 01010/10	**R/** 01010/10	**R/** 01010/10	**F/** 1011/ 22
7	Fundamental frequency of filled crypto stego container (stego + crypto)	*131*	*114*	*104*	*120*	*218*	*110*	*110*	*110*	*222*

Thus, the stego channel is to be secured from detection, and the only practical security method is a cryptography method. In the given security method normalized values of carrier features are disclosing parameters, i.e. unordinary unnatural statistical features of a filled container in comparison with an empty one. Dithering can be applied – the procedure of incorporation of a small noise signal that makes the main signal more natural, but it will complicate the decoding. The only radical solution is crypto security. It implies that in the coder output there should be QIM crypto converter that creates from standardized ones new natural-like random-like, but ciphered parameters. In the receiving end each parameter is first deciphered and then decoded by QIM in order to extract embedded data. Intruders do not obtain any information on the availability of the embedded message and cannot read it as they tap absolutely natural speech some parameters of which are modified within the limits of psychoacoustic norm with the use of crypto security.

It is to be emphasized that it presents quite a complicated procedure because auditory perception is so highly developed that it enables to identify very subtle tones of speech signal. The human ear, especially the one of an acoustic stego analyst, can easily identify features of unnatural speech. While embedding data two factors are to be taken into account: nonrandomness of signals of an empty speech container and maintenance of its quality in the process of embedding and data ciphering.

Thus, in the process of data embedding and extraction the speech signal can undergo two direct and two reverse modifications – stego modification and crypto modification. We will demonstrate that these modifications can be realized in an algorithm.

For this purpose we will use an example of data hiding in fundamental frequency modification described in [1]. The speech container “Wow… Sound editing just…” with 2 sec duration is divided into 9 segments with the fundamental frequency indicated in the first line of Table 1 (in integers Hz, the enclosure density is 1 letter per speech segment in homophonic code MTK-2 type, the stego coder interval is -0,5 ÷ +05 Hz. Long texts embedded without ciphering, for example the embedded word КОМПАНИЯ (KOMPANIYA) in line 3, can be easily detected and read by an intruder using a statistic analysis of a filled container. We will cipher this word. In flow cipher units each bit of initial information is enciphered with gamma – applying of pseudorandom figures on open data. In the given example the word EMBEDING is used as gamma with applying “exclusion OR” to each bit (XOR).

The resulting ciphered text C_SQRRRF is nore complicated for detection, but ciphering can influence the quality of the speech container (Figure 2) and arouse suspicion that stego messages are imbedded in the container.

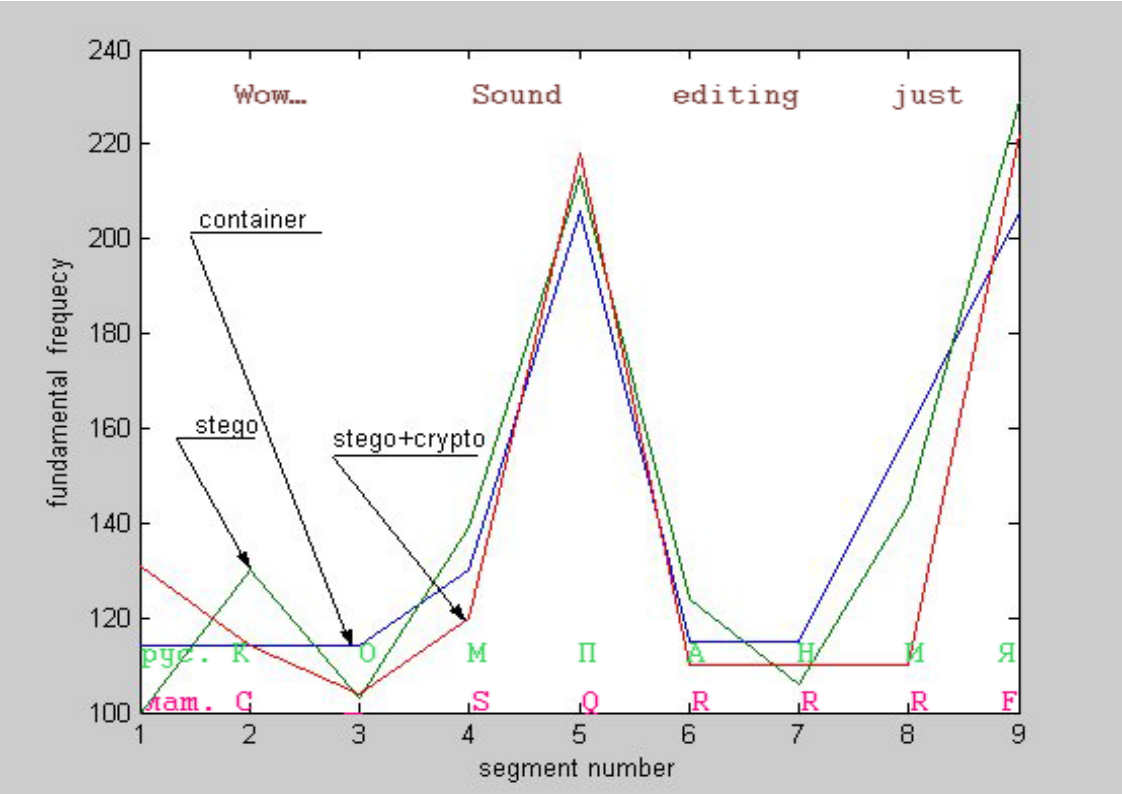


Figure 2 Example of data hiding by means of stego and crypto modification of segments fundamental frequency

III. Results

As is shown in Figure 2 cyphering does not cause destructive modifications of fundamental frequency, the melodic contour and phrase intonation are maintained. It results from the fact that the applied stego code is within the limits from 0 till 99, and the applied gamma iin MTK-2 code is from 0 till 32. This process can be called crypto flow modification analogous to dithering. The method of deciphering applying gamma repeatedly and the method of data extraction with a code table are well-known. Although the acceptable limits of fundamental frequency and other parameters modification with ciphering requires further research.

The task of data hiding with ciphering is connected with the task of synchronization of flow crypto modification that is required both at the beginning of concealed data transmission and in the process of transmission in case of temporary loss of connection. According to some authors the complexity of synchronization is an advantage from the point of view of concealed data transmission [4], that is why the use of artificial synchronization means – synchronization packages, marks, headlines etc – is undesirable. The technical complexity of synchronization task and opportunities of its realization on a real time basis are to be assessed.

IV. Conclusion

The described method of data hiding in speech signals is based on steganography with flow crypto security. In the given paper realization and crypto immunity of flow ciphers are not considered as they are subject for study of other specialists. Although stego coding combined with ciphering imposes additional distortions in the speech signal and delays in transmission. Thereby in the process of development of technical means of concealed data

transmission acceptable limits are to be defined from the perspective of security and natural speech communication.

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